

# Problem Set 2

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## Exercise 1

In the Jupyter Notebook file `ps2_ex1.ipynb`, you will find an implementation of single-agent dynamic discrete choice estimation following Rust (1987). We mostly follow the original model in the paper, with a few minor changes:

- Set of alternatives:  $J = \{1, 2\}$ :
  - Choice 1: Maintain bus engine.
  - Choice 2: Replace bus engine (which regenerates the mileage to 0).
- State variables:
  - Observable state variables,  $x_t \in \{0, \dots, X-1\}$ , are the bus mileages.
- State transition matrix:
  - Choice 1: When the engine is maintained, the transition matrix from state  $x_t$  to  $x_{t+1}$  is

$$\begin{pmatrix} \phi_1 & \phi_2 & 1 - \phi_1 - \phi_2 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \phi_1 & \phi_2 & 1 - \phi_1 - \phi_2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \phi_1 & \phi_2 & 1 - \phi_1 - \phi_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & \phi_1 & 1 - \phi_1 \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix}$$

- Choice 2: When the engine is replaced, mileage gets regenerated to 0 (i.e.,  $x_t = 0$ ) so that

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

- Discount factor,  $\beta$ , is assumed to be known.
- The deterministic component of the utility,  $u_j(x_t; \theta)$ , are parameterized as
  - Choice 1:  $u_1(x_t; \theta) = \theta_0 x_t$ .
  - Choice 2:  $u_2(x_t; \theta) = \theta_1$ .

We can interpret  $\theta_0$  as the maintenance cost and  $\theta_1$  as the replacement cost.

- Unobservable parts of the flow utility,  $\epsilon_{jt}$ , are assumed to follow a T1EV distribution with mean 0 for any  $j \in J$  and  $t$ .

**Note: we use  $(\theta_0, \theta_1)$  instead of  $(\theta_1, \theta_2)$**

1. For some different combinations of model parameters  $(\beta, \theta_0, \theta_1)$ , plot the implied true conditional choice probability of choosing to maintain the engine (i.e., Choice 1) for each state  $x_t$ . Compare your plots across different values of  $\beta$ ,  $\theta_0$ , and  $\theta_1$ . How do plots change as you vary each of those model parameters? Do the patterns you observe make sense? (Hint: Conditional choice probabilities can be calculated using `prob` from `dgp` function in the Jupyter Notebook.)

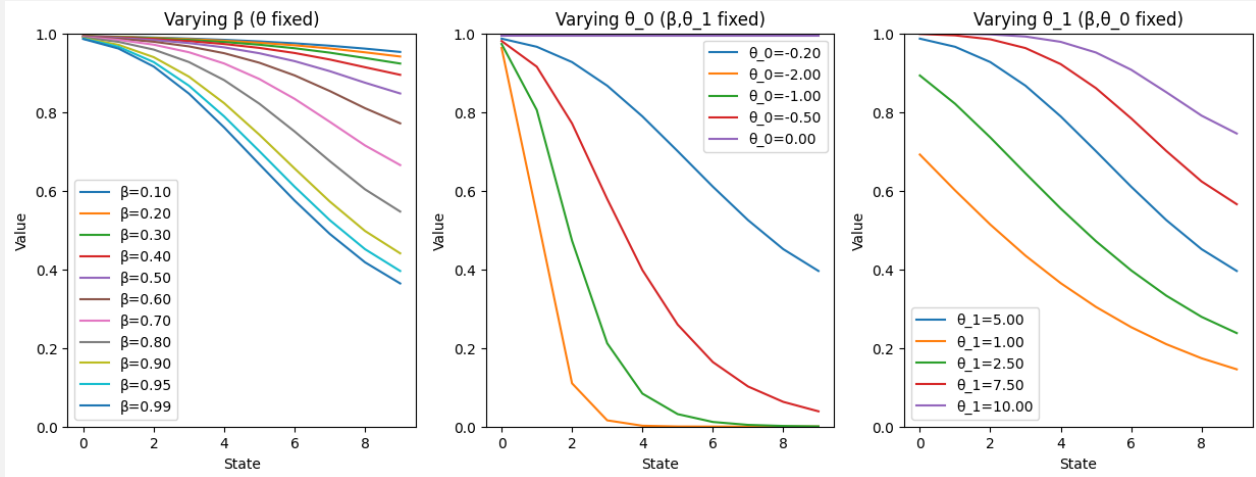


Figure 1:  $\mathbb{P}[\text{maintain}|\text{state } x]$  by  $\beta$ ,  $\theta_0$ ,  $\theta_1$

- $\beta$ : We see that as we increase how much the agent values the future (larger  $\beta$ ), they decrease the probability of maintaining the engine and increase the probability of choosing to replace it for every state possible. The intuition is that agents would prefer to pay the fixed cost now rather than incur larger costs in the future if their  $\beta$  is large.
- $\theta_0$ : We see that as the cost of maintaining the engine increases, the probability of maintaining the engine decreases and the probability of replacing the engine increases. This follows the same intuition as  $\beta$ , where if it is very costly each period to maintain an engine, it is optimal for an agent to incur the fixed cost of replacing the engine now in order to face smaller costs of maintaining the engine in the future.
- $\theta_1$ : Note that the sign convention is such that larger positive values of  $\theta_1$  correspond to larger replacement costs. We see that as the cost of replacing the engine increases, the probability of maintaining an engine increases. The intuition is that an agent would prefer to just pay the cost of maintaining the engine each period rather than paying a very expensive replacement cost.

2. We will now implement a simplified version of Hotz and Miller (1993)'s choice probability inversion theorem to estimate  $\theta_0$  and  $\theta_1$ . Let  $v_j(x_t; \theta, \phi)$  be the deterministic part of the value function for choosing choice  $j$  at state  $x_t$  so that the Bellman equation representation becomes

$$v_j(x_t; \theta, \phi) = u_j(x_t; \theta) + \beta \int \ln \left( \sum_{k \in J} \exp\{v_k(x_{t+1}; \theta, \phi)\} \right) \Pr[dx_{t+1} | x_t, a_t = j, \phi].$$

Let  $\widehat{\Pr}[a_t = j | x_t]$  and  $\hat{v}_j(x_t)$  be the sample analogue of  $\Pr[a_t = j | x_t; \theta, \phi]$  and  $v_j(x_t; \theta, \phi)$  respectively. Then, the

sample analogue of the conditional choice probability can be written as

$$\widehat{\Pr}[a_t = j \mid x_t] = \frac{\exp(\hat{v}_j(x_t))}{\sum_{k \in J} \exp(\hat{v}_k(x_t))}.$$

Let  $J$  be the reference choice, or outside option, in which we normalize  $v_J(x_t; \theta, \phi) = 0$ . Then, using the above sample analogue representation, we can write

$$\ln \left( \frac{\widehat{\Pr}(a_t = j \mid x_t)}{\widehat{\Pr}(a_t = J \mid x_t)} \right) = \hat{v}_j(x_t) - \hat{v}_J(x_t) = \underbrace{\hat{v}_j(x_t) - \hat{v}_J(x_t)}_{=0 \text{ by normalization}}.$$

As a result, the maximum-likelihood estimator would proceed with the objective function

$$\sum_t \sum_{j \in J} \mathbf{1}(a_t = j \mid x_t) \log \left( \frac{\exp(\hat{v}_j(x_t; \theta, \phi))}{\sum_{k \in J} \exp(\hat{v}_k(x_t; \theta, \phi))} \right).$$

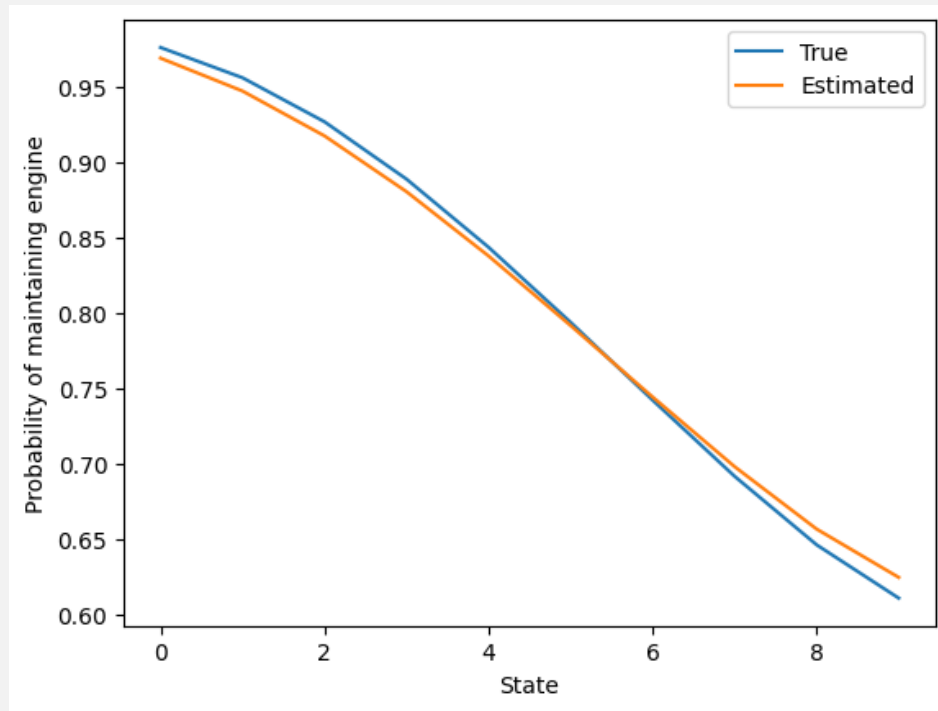
Complete the functions `csvf_from_ccp`, `likelihood_two_step_ccp`, and `two_step_ccp` under “Estimation 2: Conditional choice probability inversion” section of the Jupyter Notebook to implement the estimation method illustrated above. (The suggested flow of implementation in the Jupyter Notebook is there only to help you. Therefore, you are free to implement the estimation method above with your own functions instead of completing the functions in the Jupyter Notebook if you would prefer.)

Completed in Code Portion

3. Choose your true model parameters and simulate data using `draw` function in the Jupyter Notebook. Use the choice probability inversion method you implemented in Part 2 to estimate  $\theta = (\theta_0, \theta_1)$  and plot the implied conditional choice probabilities. How do they compare to your true values of  $\theta$  and the corresponding true conditional choice probabilities?

|           | $\theta_0$ | $\theta_1$ |
|-----------|------------|------------|
| True      | -0.1       | 4          |
| Estimated | -0.0910    | 3.70       |

As shown, the choice probability inversion returns estimates that are  $\pm 10\%$  the true values of  $\theta_0$  and  $\theta_1$ . This results in estimated conditional choice probabilities (shown below) that match the true conditional choice probabilities quite closely. We show in Part 4 that the other methods (NXFP, MPEC, and NPL) are more accurate, but the choice probability inversion method seems fairly good given that it tends to be faster than these other methods.

Figure 2:  $\mathbb{P}[\text{maintain}|\text{state } x], \theta$  vs.  $\hat{\theta}_{ccp}$ 

4. Repeat the Part 3 using (i) nested fixed-point algorithm, (ii) nested pseudo-likelihood, and (iii) MPEC (all of them are already coded-up in the Jupyter Notebook). Compare their speed and accuracy for different values of the model parameters and discuss.

Table 1: Estimation Accuracy Across Methods

| Method                   | Avg. % $\theta_0$ error |   | Avg. % $\theta_1$ error |   |
|--------------------------|-------------------------|---|-------------------------|---|
| MPEC                     | 7.92                    | % | 3.76                    | % |
| Nested fixed point       | 7.92                    | % | 3.76                    | % |
| Nested pseudo-likelihood | 7.92                    | % | 3.76                    | % |
| CCP inversion            | 9.99                    | % | 4.75                    | % |

Table 2: Average Estimation Time

| Method                   | Time (seconds) |
|--------------------------|----------------|
| MPEC                     | 0.022          |
| Nested fixed point       | 0.271          |
| Nested pseudo-likelihood | 0.199          |
| CCP inversion            | 0.011          |

As shown in Table 1 and Figure 3, MPEC, the nexted fixed-point algorithm (NXFP), and nested pseudo likelihood (NPL) are quite accurate and produce remarkably similar estimates and implied conditional choice probabilities. In fact, the estimated choice probabilities from NPL, NXFP, and MPEC are indistinguishable in Figure 3. After using claude code to speed up and vectorize operations, average estimation times for all 4

methods are very fast. Finally, conditional choice probability (CCP) inversion is in general less accurate, at the benefit of being around 50% faster than the NXFP.

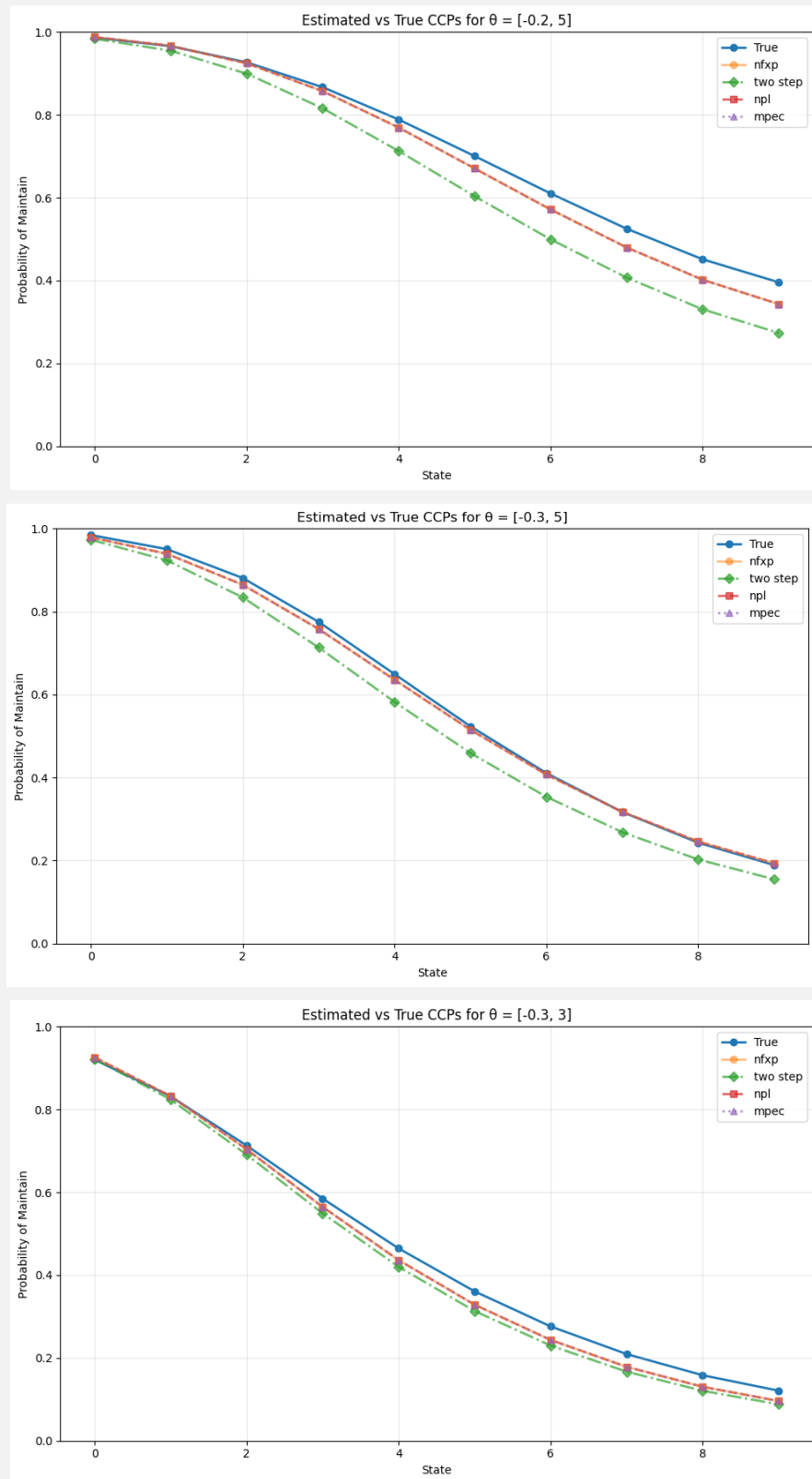


Figure 3:  $\mathbb{P}[\text{maintain}|\text{state } x]$ , by method  
(nfxp, mpec, npl overlap)

## Exercise 2

In the file `ps2_ex2.csv`, you will find mileage data for the bus managed by Harold Zurcher. For this exercise, there are no covariates. In each period, Harold Zurcher chooses whether to perform maintenance on the bus or to replace the engine. Flow utility is given by

$$u(x_t, d_t) + \epsilon_{a,t} = \begin{cases} -\theta_1 x_t - \theta_2 \left(\frac{x_t}{100}\right)^2 + \epsilon_{0,t} & \text{if } d_t = 0 \\ -\theta_3 + \epsilon_{1,t} & \text{if } d_t = 1 \end{cases}$$

where  $x_t$  is the current mileage of the bus,  $d_t$  is the choice of Harold Zurcher, and  $\theta$  is a vector of parameters. The unobserved utility terms  $\epsilon_{a,t}$  are independently distributed Type I Extreme Value.

Harold Zurcher maximizes his lifetime discounted utility with discount factor  $\beta$ . The state variable evolves according to the transition probability

$$p(x_{t+1} | x_t, d_t) = \begin{cases} g(x_{t+1} - 0) & \text{if } d_t = 1 \\ g(x_{t+1} - x_t) & \text{if } d_t = 0 \end{cases}$$

That is, replacing the engine regenerates mileage to zero.

1. How can you recover engine replacement decisions from the mileage data? Store these decisions in a separate variable.

Since mileage increases if one does not replace the engine, a decrease in mileage from time  $t$  to  $t+1$  indicates an engine replacement in time  $t$ .

2. Discuss the conditional independence assumption.

Given that  $\epsilon_{a,t}$  are independently distributed *T1EV*, the conditional independence assumption is the following:

$$\mathbb{P}(x_{t+1}, \epsilon_{t+1} | x_t, d_t) = q(\epsilon_{t+1})p(x_{t+1} | x_t, d_t)$$

where  $q(\epsilon_{t+1})$  is the pdf of  $\epsilon_{t+1}$ . In other words, the distribution of the state variable  $x_{t+1}$ , conditional on the previous state  $x_t$  and previous decision  $d_t$ , is independent of the distribution of the unobserved utility shock  $\epsilon_{t+1}$ . Now,  $\epsilon$  influences  $x_{t+1}$ , only through  $\epsilon_t$ 's impact on  $d_t$ . This assumption greatly reduces the complexity of our problem. Instead of having to estimate and keep track of the joint probability of  $(x_{t+1}, \epsilon_{t+1})$ , we can simply use their individual distributions. Then, since  $\epsilon_t$  is independent and T1EV, we can easily integrate over it's distribution, meaning that we now only have to concern ourselves with estimating  $p(x_{t+1} | x_t, d_t)$  and do not have to keep track of the  $\{\epsilon_t\}$  state variables in our estimation procedure.

3. Discretize the domain of  $x_t$  into  $K$  bins. Estimate the Markov transition probabilities

$$p(x_{t+1} | x_t, d_t)$$

This should result in two  $K \times K$  stochastic matrices, one for each action  $d_t$ . Choose  $K$  appropriately.

We choose  $K = 10$ , i.e. we discretized mileage into 10 bins. In general, we can see that conditional on maintaining the engine, there is around a 55 – 65% probability of remaining in the current state and a 45 – 35% chance of transitioning to the next highest state. As predicted, when choosing to replace the engine, there is a  $\approx 99\%$  chance of returning to state 0 and  $\approx 1\%$  chance of moving to state 1.

Table 3: Transition matrix for maintaining engine

|   | 0     | 1     | 2       | 3       | 4     | 5       | 6       | 7     | 8     | 9     |
|---|-------|-------|---------|---------|-------|---------|---------|-------|-------|-------|
| 0 | 0.473 | 0.526 | 0.00131 | 0       | 0     | 0       | 0       | 0     | 0     | 0     |
| 1 | 0     | 0.557 | 0.442   | 0.00109 | 0     | 0       | 0       | 0     | 0     | 0     |
| 2 | 0     | 0     | 0.554   | 0.446   | 0     | 0       | 0       | 0     | 0     | 0     |
| 3 | 0     | 0     | 0       | 0.561   | 0.438 | 0.00126 | 0       | 0     | 0     | 0     |
| 4 | 0     | 0     | 0       | 0       | 0.587 | 0.411   | 0.00165 | 0     | 0     | 0     |
| 5 | 0     | 0     | 0       | 0       | 0     | 0.599   | 0.401   | 0     | 0     | 0     |
| 6 | 0     | 0     | 0       | 0       | 0     | 0       | 0.670   | 0.330 | 0     | 0     |
| 7 | 0     | 0     | 0       | 0       | 0     | 0       | 0       | 0.696 | 0.304 | 0     |
| 8 | 0     | 0     | 0       | 0       | 0     | 0       | 0       | 0     | 0.571 | 0.429 |
| 9 | 0     | 0     | 0       | 0       | 0     | 0       | 0       | 0     | 0     | 1     |

Table 4: Transition matrix for replacing engine

|   | 0     | 1       | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|-------|---------|---|---|---|---|---|---|---|---|
| 0 | 0     | 0       | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1     | 0       | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 1     | 0       | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0.977 | 0.0227  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0.990 | 0.0103  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 0.990 | 0.00980 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | 0.989 | 0.0112  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 7 | 1     | 0       | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 8 | 1     | 0       | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 9 | 1     | 0       | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

4. Define the expected value function

$$EV(x, d) = \int V_\theta(y, \epsilon) p(d\epsilon) p(dy \mid x, d)$$

Show that the following fixed point equation holds:

$$EV(x, d) = \int \log \left( \sum_j \exp(u(y, j) + \beta EV(y, j)) \right) p(dy \mid x, d)$$

Note: I drop the  $\theta$  argument for ease of notation, but I include it in the next parts.

Assuming the regularity conditions for stochastic dynamic programming are satisfied, we can represent the agents problem at a given time-period for a given set of parameters  $\theta$  governing their utility as:

$$V(x_t, \epsilon_t) = \max_{d_t \in \{0,1\}} \{u(x_t, d_t) + \epsilon_{a,t} + \beta EV(x_{t+1}, \epsilon_{t+1} \mid x_t, \epsilon_t, d_t)\}$$

Then, by the conditional independence assumption, the fact that the transition process is first-order markov, and the distributional assumptions on  $\epsilon_t$ , we can write the expected value of this value function as follows:

$$\mathbb{E}[V(x, d)] = \int_y \int_{\epsilon} \max_{d \in \{0,1\}} \{u(y, d) + \epsilon_{a,t} + \beta \mathbb{E}V(y, \epsilon_{t+1} \mid x, \epsilon_t, d)\} p(d\epsilon) p(dy \mid x, d)$$

where for ease of notation we have denoted  $d = d_t, x = x_t, y = x_{t+1}, p(y|x, d)dx = p(dy|x, d)$  and  $p(\epsilon)d\epsilon = p(d\epsilon)$ . Then, using that when  $u_{i,j} = \delta_j + \epsilon_{i,j}$  and  $\epsilon_{i,j}$  is *T1EV*,  $\mathbb{E}[\max_{j \in \mathcal{J}} \{u_{i,j}\}] = \ln[\sum_{j \in \mathcal{J}} \exp(\delta_j)] + \gamma$ , we have that

$$\begin{aligned} \mathbb{E}V(x, d) &= \int \left( \ln \left[ \sum_{j=0,1} \exp(u(y, j) + \beta \mathbb{E}V(y, j)) \right] + \gamma \right) p(dy \mid x, d) \\ &= \int \ln \left[ \sum_{j=0,1} \exp(u(y, j) + \beta \mathbb{E}V(y, j)) \right] p(dy \mid x, d) \quad \text{assuming } \mathbb{E}[\epsilon_t] = 0 \end{aligned}$$

This means that we can solve for  $\mathbb{E}V(x, d)$ , and therefore  $V(x, d)$ , using a fixed-point algorithm.

5. Derive the conditional choice probabilities using  $EV(x, d)$  and  $\theta$ .

The likelihood of an individual choosing  $d_t = j$ , given state  $x_t$  and utility parameters  $\theta$ , is:

$$\begin{aligned} \mathbb{P}(d_t = 1 \mid x_t; \theta) &= \mathbb{P}(u(x_t, 1; \theta) + \beta \mathbb{E}V(x_{t+1}, \epsilon_{t+1} \mid x_t, 1; \theta) + \epsilon_t(1) > u(x_t, 0; \theta) + \beta \mathbb{E}V(x_{t+1}, \epsilon_{t+1} \mid x_t, 0; \theta) + \epsilon_t(0)) \\ &= \frac{\exp(u(x_t, 1; \theta) + \beta \mathbb{E}V(x_{t+1}, \epsilon_{t+1} \mid x_t, 1; \theta))}{\sum_{j=0,1} \exp(u(x_t, d_t = j; \theta) + \beta \mathbb{E}V(x_{t+1}, \epsilon_{t+1} \mid x_t, d_t = j; \theta))} \end{aligned}$$

and likewise

$$\mathbb{P}(d_t = 0 \mid x_t; \theta) = \frac{\exp(u(x_t, 0; \theta) + \beta \mathbb{E}V(x_{t+1}, \epsilon_{t+1} \mid x_t, 0; \theta))}{\sum_{j=0,1} \exp(u(x_t, d_t = j; \theta) + \beta \mathbb{E}V(x_{t+1}, \epsilon_{t+1} \mid x_t, d_t = j; \theta))}$$

where we have used the formula for the difference in two independent *T1EV* RVs.

6. Reduce the state space of  $EV$  using the regenerative property.

In this setting, replacing the engine resets the state in the next period to the first state  $x = 1$ . As a result, when considering the expected continuation value of performing  $j = 1$ , we know that is equal to  $w(x = 1, \theta)$  rather than some integral over  $w(x, \theta)$ . Therefore, the only relevant states for the replacement decision are the current state and the next period state of  $x = 1$ , rather than all current and all next-period states. This reduces the state space, as shown formally in the aside to part 7.

7. Rewrite the fixed point equation as a matrix equation.

Let's consider the deterministic portion of the choice-specific value function for choices  $j = 0, 1$ :

$$\begin{aligned} v_j(x_t; \theta) &\equiv v_j(x_t, \epsilon_t; \theta) - \epsilon_t(j) \\ &= u(x_t, d_t = j; \theta) + \beta \mathbb{E}V(x_{t+1}, \epsilon_{t+1} \mid x_t, d_t = j; \theta) \end{aligned}$$



Then, we can use the expression for  $\mathbb{E}V(x, d)$  found in part 4 to rewrite this function as the following fixed point equation:

$$\begin{aligned} v_j(x_t; \theta) &= u(x_t, d_t = j; \theta) + \beta \int \ln \left[ \sum_{j=0,1} \exp(u(y, j) + \beta \mathbb{E}V(y, j)) \right] p(dy \mid x, d) \\ &= u(x_t, d_t = j; \theta) + \beta \int \ln \left[ \sum_{j=0,1} \exp(v_j(y; \theta)) \right] p(dy \mid x, d) \end{aligned}$$

This equation implicitly defines  $v_j(x_t; \theta)$  for any choice  $j$  and state  $x_t$ . To re-write this in matrix form, notice that since the state follows a first-order markov process with discrete states  $k = 1, \dots, K \in \mathcal{X}$ :

$$\int \ln \left[ \sum_{k=0,1} \exp(v_k(y; \theta)) \right] p(dy \mid x, d) = \sum_{k \in \mathcal{X}} \underbrace{\ln \left[ \sum_{j=0,1} \exp(v_j(y; \theta)) \right]}_{\equiv w(y; \theta)} P_{xk|j}$$

where  $P_{xk|j}$  is the  $(x, k)$  entry in the Markov transition matrix given  $d_t = j$  and gives the probability of transitioning from state  $x \in \mathcal{X}$  to  $k \in \mathcal{X}$  conditional on  $j$  having been chosen at time  $t$ . Denoting the conditional Markov transition matrix as  $M(j)$ , notice that

$$M(j) \begin{bmatrix} w(1; \theta) \\ \vdots \\ w(K; \theta) \end{bmatrix} = \begin{bmatrix} \sum_{k \in \mathcal{X}} P_{xk|j} w(k; \theta) = \mathbb{E}[w(x_{t+1}, ; \theta) \mid x = 1] \\ \vdots \\ \mathbb{E}[w(x_{t+1}, ; \theta) \mid x = K] \end{bmatrix}$$

and so it follows that

$$\begin{bmatrix} v_j(1; \theta) \\ \vdots \\ v_j(K; \theta) \end{bmatrix} = \begin{bmatrix} u(1, d_t = j; \theta) \\ \vdots \\ u(K, d_t = j; \theta) \end{bmatrix} + \beta M(j) \begin{bmatrix} w(1; \theta) \\ \vdots \\ w(K; \theta) \end{bmatrix}$$

where  $w(y; \theta) = \ln \left[ \sum_{j=0,1} \exp(v_j(y; \theta)) \right]$  and  $M(d)$  is a  $K \times K$  markov matrix. If we stack this equation for  $j = 0, 1$  and define  $\mathcal{M} = \begin{bmatrix} M(0) \\ M(1) \end{bmatrix}$ , we can fully characterize this fixed point equation as a matrix equation:

$$\begin{bmatrix} v_0(1; \theta) \\ \vdots \\ v_0(K; \theta) \\ v_1(1; \theta) \\ \vdots \\ v_1(K; \theta) \end{bmatrix} = \begin{bmatrix} u(1, d_t = 0; \theta) \\ \vdots \\ u(K, d_t = 0; \theta) \\ u(1, d_t = 1; \theta) \\ \vdots \\ u(K, d_t = 1; \theta) \end{bmatrix} + \beta \mathcal{M} \begin{bmatrix} w(1; \theta) \\ \vdots \\ w(K; \theta) \end{bmatrix}$$

**Aside on regenerative property:** Let  $j = 1$  denote replacing an engine. Then  $M(1) = \begin{bmatrix} 1 & 0 & \dots \\ \vdots & \vdots & \dots \\ 1 & 0 & \dots \end{bmatrix}$  and

so

$$\begin{bmatrix} v_1(1; \theta) \\ \vdots \\ v_1(K; \theta) \end{bmatrix} = \begin{bmatrix} u(1, d_t = 1; \theta) \\ \vdots \\ u(K, d_t = 1; \theta) \end{bmatrix} + \beta \begin{bmatrix} w(1; \theta) \\ \vdots \\ w(1; \theta) \end{bmatrix}$$

8. Write a function that solves the fixed point equation using Rust's algorithm.

Completed in the code

9. Write a function that computes the likelihood of the sample for any parameter  $\theta$ .

Completed in the code

10. Estimate  $\theta$  using  $\beta = 0.999$ .

|                | $\hat{\theta}_1$ | $\hat{\theta}_2$ | $\hat{\theta}_3$ |
|----------------|------------------|------------------|------------------|
| NFXP Estimates | -.240            | 479              | 4.19             |

The estimated parameter values result in the following estimated utility function:

$$u(x_t, d_t) + \epsilon_{a,t} = \begin{cases} .24x_t - .0479x_t^2 + \epsilon_{0,t} & \text{if } d_t = 0 \\ -4.19 + \epsilon_{1,t} & \text{if } d_t = 1 \end{cases}$$

This is a somewhat surprising result as it suggest that up to around state 3, the cost of maintaining an engine is *decreasing* in mileage. After which, the cost of maintaining an engine increases in the mileage, as is expected. Nevertheless, Figure 4 shows that these estimated parameters result in implied conditional maintenance probabilities that are fairly similar to those observed in the data. However, we can see that in the data, the likelihood of replacing an engine increases slightly from state 7 to state 8, which is consistent with the cost of maintaining an engine decreasing over these states. This increase may be driven by there being unmodeled idiosyncratic or heterogenous utility shocks in the data, meaning that our model is misspecified or that our assumption of *i.i.d* errors is not satisfied. As a result, our estimates of  $\theta_1$  and  $\theta_2$  may be biased upward and downward, respectively, in order to rationalize this "blip" in the data. Another reason for our  $\theta$ 's implying a decreasing cost of maintenance in early states may be due to the face that there exists learning by doing. For example, in states 0 and 1, the truck's engine was just recently replaced, meaning that the mechanics might need to expend extra effort to familiarize themselves with the new engine and other replaced parts. However, after a few periods, the mechanics are more familiar with the engine/new parts, making maintenance relatively less costly.

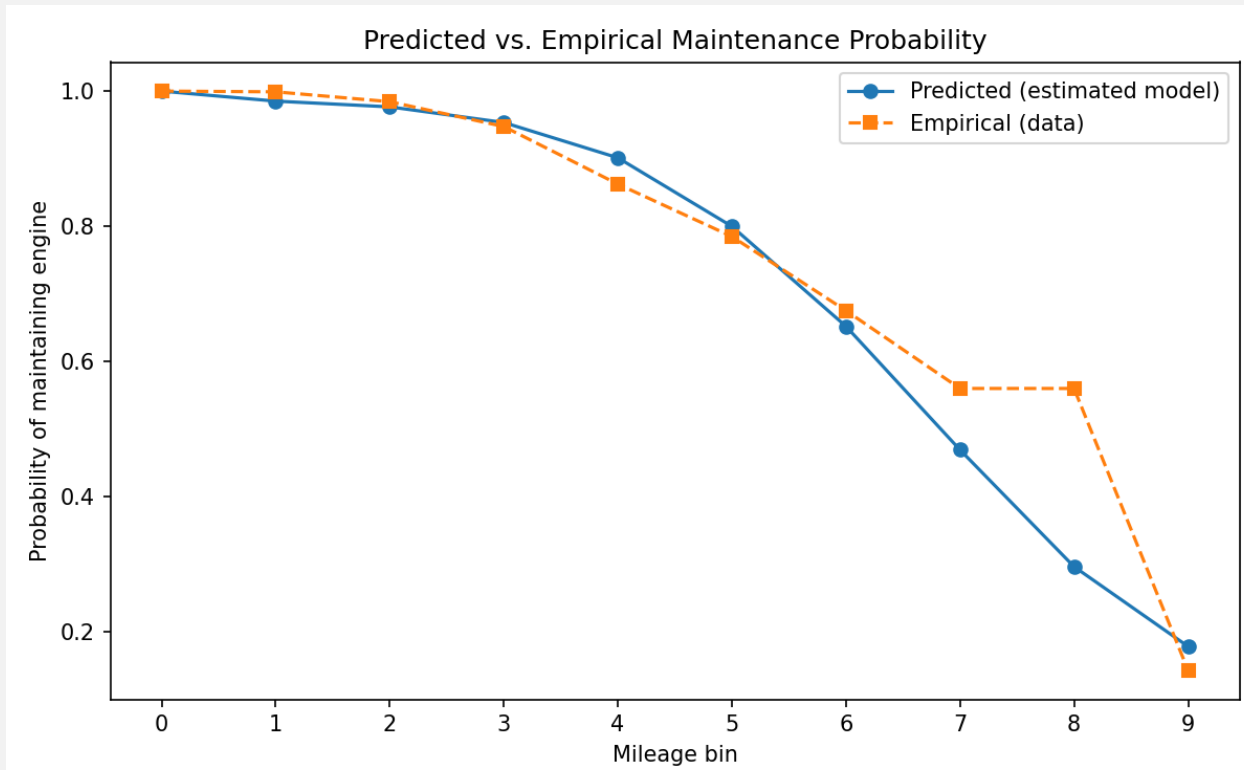


Figure 4: Predicted vs. observed maintenance probabilities.

### Exercise 3

Consider a static entry game between  $J$  firms operating in  $T$  markets. Upon entry, the profit of firm  $j$  in market  $t$  depends on the number of entrants and is given by:

$$\pi_{jt}(n_{-jt}) = x_{jt}\beta - \phi_j - \delta_j \log(1 + n_{-jt}) + \epsilon_{jt}$$

where  $x_{jt}$  are market-firm specific profit shifters,  $\phi_j$  is a firm specific fixed cost,  $n_{-jt}$  is the number of competitors in market  $t$  and  $\epsilon_{jt} \sim F$  is an idiosyncratic shock. Given  $n_{-jt}$  firm  $j$  enters if and only if  $\pi_{jt} \geq 0$ .

1. For a given market  $t$ , find the pure strategy Nash Equilibrium (or Equilibria) as a function of the realization of  $(\epsilon_{1t}, \dots, \epsilon_{Jt})$ . For simplicity assume that  $J = 2$ .

Note: I assume  $\delta_j > 0$  for the entirety of the exercise

For 2 firms in market  $t$ , this game can be represented by the following payoff matrix:

|       | $O_2$                                     | $E_2$                                                                                                            |
|-------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| $O_1$ | 0, 0                                      | $0, x_{2t}\beta - \phi_2 + \epsilon_{2t}$                                                                        |
| $E_1$ | $x_{1t}\beta - \phi_1 + \epsilon_{1t}, 0$ | $x_{1t}\beta - \phi_1 - \delta_1 \ln(2) + \epsilon_{1t}, x_{2t}\beta - \phi_2 - \delta_2 \ln(2) + \epsilon_{2t}$ |

where  $E_j, O_j$  represent firm  $j$ 's decision to enter or not enter the market, respectively. The pure-strategy

Nash Equilibria are as follows:

$$\begin{aligned}
 (O_1, O_2) \quad & \text{if } x_{1t}\beta - \phi_1 + \epsilon_{1t} < 0 \quad \text{and} \quad x_{2t}\beta - \phi_2 + \epsilon_{2t} < 0 \\
 (E_1, E_2) \quad & \text{if } x_{1t}\beta - \phi_1 - \delta_1 \ln(2) + \epsilon_{1t} > 0 \quad \text{and} \quad x_{2t}\beta - \phi_2 - \delta_2 \ln(2) + \epsilon_{2t} > 0 \\
 (O_1, E_2) \quad & \text{if } x_{1t}\beta - \phi_1 - \delta_1 \ln(2) + \epsilon_{1t} < 0 \quad \text{and} \quad x_{2t}\beta - \phi_2 + \epsilon_{2t} > 0 \\
 (E_1, O_2) \quad & \text{if } x_{1t}\beta - \phi_1 + \epsilon_{1t} > 0 \quad \text{and} \quad x_{2t}\beta - \phi_2 - \delta_2 \ln(2) + \epsilon_{2t} < 0
 \end{aligned}$$

2. Suppose you have data on the entry decisions of both firms and some relevant characteristics  $\{y_{1t}, y_{2t}, x_{1t}, x_{2t}\}_{t=1}^T$ , where  $y_{jt} = 1$  if firm  $j$  entered market  $t$ . Write the likelihood of the data as  $\prod_{t=1}^T l(y_{1t}, y_{2t} \mid x_{1t}, x_{2t}, \beta, \delta_1, \delta_2)$ , where for each  $t$ ,  $(y_{1t}, y_{2t})$  is a Nash Equilibrium. Is the likelihood well-defined? Explain.

The likelihood of observing a sequence of Nash equilibria  $\{(y_{1t}, y_{2t})\}_{t=1}^T$ , given  $x_{1t}, x_{2t}$  and assuming that  $\epsilon_{jt}$  are independent across markets, is given by:

$$\begin{aligned}
 \mathcal{L}(y_{1t}, y_{2t}) = \prod_{t=1}^T & \left( \mathbb{1}\{(0, 0)\} \mathbb{P}(\epsilon_{1t} < \phi_1 - x_{1t}\beta, \epsilon_{2t} < \phi_2 - x_{2t}\beta) \right. \\
 & + \mathbb{1}\{(0, 1)\} \mathbb{P}(\epsilon_{1t} < \phi_1 + \delta_1 \ln(2) - x_{1t}\beta, \epsilon_{2t} > \phi_2 - x_{2t}\beta) \\
 & + \mathbb{1}\{(1, 0)\} \mathbb{P}(\epsilon_{1t} > \phi_1 - x_{1t}\beta, \epsilon_{2t} < \phi_2 + \delta_2 \ln(2) - x_{2t}\beta) \\
 & \left. + \mathbb{1}\{(1, 1)\} \mathbb{P}(\epsilon_{1t} > \phi_1 + \delta_1 \ln(2) - x_{1t}\beta, \epsilon_{2t} > \phi_2 + \delta_2 \ln(2) - x_{2t}\beta) \right)
 \end{aligned}$$

where  $\mathbb{1}\{(a, b)\}$  is an indicator for observing  $y_{1t} = a, y_{2t} = b$ .

This likelihood is not well-defined as it is greater than 1, i.e. there are multiple equilibria that can arise from a given realization of  $x_{1t}, x_{2t}, \epsilon_{1t}, \epsilon_{2t}$ . To see this note that if:

$$\phi_1 - x_{1t}\beta < \epsilon_{1t} < \phi_1 + \delta_1 \ln(2) - x_{1t}\beta$$

and

$$\phi_2 - x_{2t}\beta < \epsilon_{2t} < \phi_2 + \delta_2 \ln(2) - x_{2t}\beta$$

both outcomes  $(1, 0)$  and  $(0, 1)$  can be realized. If we ignore this overlap, then the probabilities in the likelihood function will add up to greater than 1.

3. Now suppose that entry is sequential and assume that firm 1 moves before firm 2. Let  $\prod_{t=1}^T l(y_{1t}, y_{2t} \mid x_{1t}, x_{2t}, \beta, \delta_1, \delta_2)$  the likelihood of the data where, for each  $t$ ,  $(y_{1t}, y_{2t})$  is a Subgame Perfect Nash Equilibrium. Is the likelihood well-defined? Explain.

Suppose that entry is sequential, and firm 1 moves first, and the following distribution is realized:

$$\phi_1 - x_{1t}\beta < \epsilon_{1t} < \phi_1 + \delta_1 \ln(2) - x_{1t}\beta$$

and

$$\phi_2 - x_{2t}\beta < \epsilon_{2t} < \phi_2 + \delta_2 \ln(2) - x_{2t}\beta$$

then *only* firm 1 will enter. Intuitively, under these parameter values, firm 1 knows that if they enter, they will deter firm 2 from entering, and so they will make a profit. The likelihood function is now given by:

$$\begin{aligned} \mathcal{L}(y_{1t}, y_{2t}) = & \prod_{t=1}^T \left( \mathbb{1}\{(0, 0)\} \mathbb{P}(\epsilon_{1t} < \phi_1 - x_{1t}\beta, \epsilon_{2t} < \phi_2 - x_{2t}\beta) \right. \\ & + \mathbb{1}\{(0, 1)\} \left( \mathbb{P}(\phi_1 - x_{1t}\beta < \epsilon_{1t} < \phi_1 + \delta_1 \ln(2) - x_{1t}\beta, \epsilon_{2t} > \phi_2 + \delta_2 \ln(2) - x_{2t}\beta) \right. \\ & \left. + \mathbb{P}(\epsilon_{1t} < \phi_1 - x_{1t}\beta, \epsilon_{2t} > \phi_2 - x_{2t}\beta) \right) \\ & + \mathbb{1}\{(1, 0)\} \mathbb{P}(\epsilon_{1t} > \phi_1 - x_{1t}\beta, \epsilon_{2t} < \phi_2 + \delta_2 \ln(2) - x_{2t}\beta) \\ & \left. + \mathbb{1}\{(1, 1)\} \mathbb{P}(\epsilon_{1t} > \phi_1 + \delta_1 \ln(2) - x_{1t}\beta, \epsilon_{2t} > \phi_2 + \delta_2 \ln(2) - x_{2t}\beta) \right) \end{aligned}$$

which is well defined as we no longer have a set of parameter values that are consistent with multiple equilibria, so they probabilities should sum to 1.

4. Assume we have market level data on  $T$  markets. For each market  $t$ , we only observe the number of firms  $n_t$  and some market level characteristic  $x_t$ . Moreover, assume that firms are homogeneous and play a sequential entry game in each market with the same entry profits as before:

$$\pi_t(n_t) = x_t\beta - \phi - \delta \ln(n_t) + \epsilon_t.$$

- (a) What are the conditions under which  $n_t$  is a Nash Equilibrium?

This case is the “original” solution to the incompleteness problem – we consider the likelihood of observing  $n$  entrants. Assuming there are  $J$  firms in each market, we can write the following conditions under which  $n_t$  is a Nash Equilibrium:

$$n_t = 0 \quad \text{if} \quad \epsilon_t < \phi - x_t\beta$$

$$n_t = n \in [1, J-1] \quad \text{if} \quad \phi + \delta \ln(n) - x_t\beta \leq \epsilon_t < \phi + \delta \ln(n+1) - x_t\beta$$

$$n_t = J \quad \text{if} \quad \epsilon_t \geq \phi + \delta \ln(J) - x_t\beta$$

which results in the following well-defined likelihood function for  $j = 2$  (which we can generalize to  $J$  firms) :

$$\begin{aligned} \mathcal{L}(n_t) = & \prod_{t=1}^T \left( \mathbb{1}\{n_t = 0\} \mathbb{P}(\epsilon_t < \phi - x_t\beta) \right. \\ & + \mathbb{1}\{n_t = 1\} (\mathbb{P}(\phi - x_t\beta \leq \epsilon_t < \phi + \delta \ln(2) - x_t\beta)) \\ & \left. + \mathbb{1}\{n_t = 2\} \mathbb{P}(\epsilon_{1t} \geq \phi + \delta \ln(2) - x_t\beta) \right) \end{aligned}$$

- (b) Assuming that  $\epsilon_t \stackrel{i.i.d}{\sim} N(0, 1)$ , use the data in ps2\_ex3.csv to estimate the parameters  $(\beta, \phi, \delta)$  using MLE.

The estimated parameter values are:

|               | $\hat{\beta}$ | $\hat{\phi}$ | $\hat{\delta}$ |
|---------------|---------------|--------------|----------------|
| MLE Estimates | 2.15          | 1.65         | 10.51          |

which fits the data quite well, as shown in the figure below. Intuitively, higher values of  $x_t$  (say, product quality) result in greater profits, with there being a fixed cost to entry  $\phi$  and profit in a market decreasing (at a decreasing rate) in the number of other firms in the market.

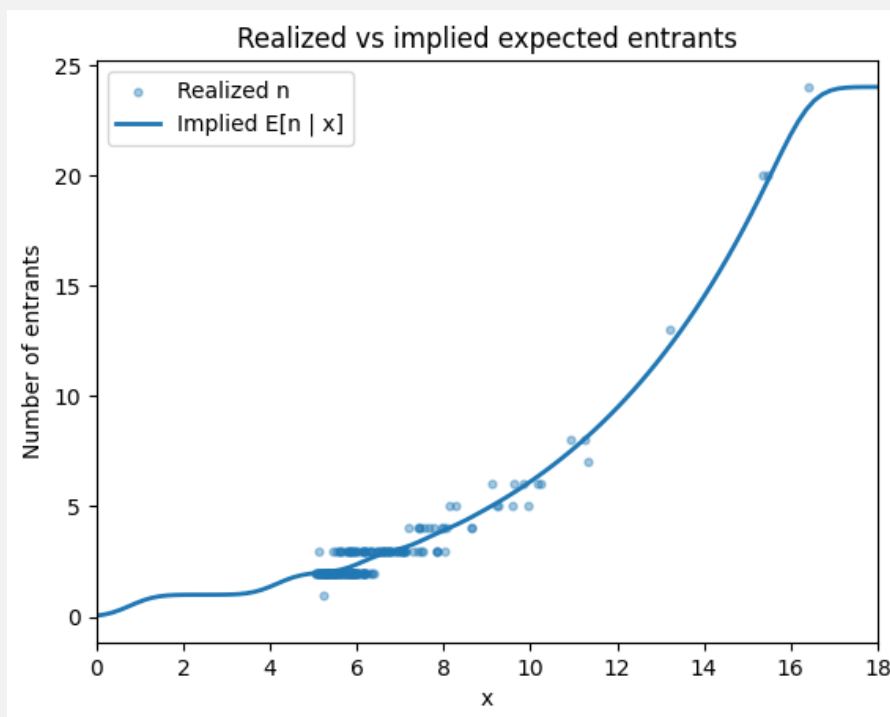


Figure 5: Expected number of entrants, data vs estimated